

Foundation (Jalapeno)

Q1. If $2^x = 16$, what is x ?

- (A) 2
- (B) 3
- (C) 4
- (D) 5
- (E) 6

Q2. Solve for y in $y = 3^x$ when $y = 27$

- (A) 1
- (B) 2
- (C) 3
- (D) 4
- (E) 5

Q3. What is the value of x in $5^x = 125$?

- (A) 2
- (B) 3
- (C) 4
- (D) 5
- (E) 6

Q4. A space station is traveling at a speed of 2^x km/h. If the speed is 32 km/h, what is x ?

- (A) 2
- (B) 3
- (C) 4
- (D) 5
- (E) 6

Q5. A scientist is studying the growth of bacteria in a lab. The

population of bacteria is given by 3^x . If the population is 81, what is x ?

- (A) 2
- (B) 3
- (C) 4
- (D) 5
- (E) 6

Q6. Solve for x in $x = \log_2(64)$

- (A) 4
- (B) 5
- (C) 6
- (D) 7
- (E) 8

Q7. A planet is orbiting the sun at a distance of 4^x km. If the distance is 256 km, what is x ?

- (A) 2
- (B) 3
- (C) 4
- (D) 5
- (E) 6

Q8. What is the value of x in $2^{2x} = 256$?

- (A) 2
- (B) 3
- (C) 4
- (D) 5
- (E) 6

Open Ended Questions (FRQ)

Q9. Solve for y in $y = 2^x$ when $y = 128$. Show your work and provide the final answer.

Q10. A spacecraft is traveling through space with a velocity of 3^x m/s. If the velocity is 243 m/s, what is x ? Provide your solution and final answer.

Chef's Tips

- Read each question carefully
- Use a calculator only when necessary
- Show all work for free response questions